



Dwight Look College of

ENGINEERING

TEXAS A&M UNIVERSITY

Team 64: TI Solar Battery Pack Bi-Weekly Update 3

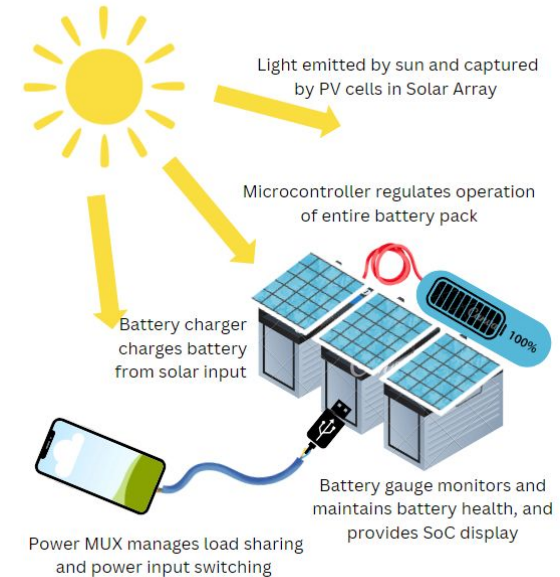
Jaein Cha, Patrick Wang, Cole Miller, Camden Pomeroy

Sponsor: Wyatt Keller

TA: Ali Alenezi

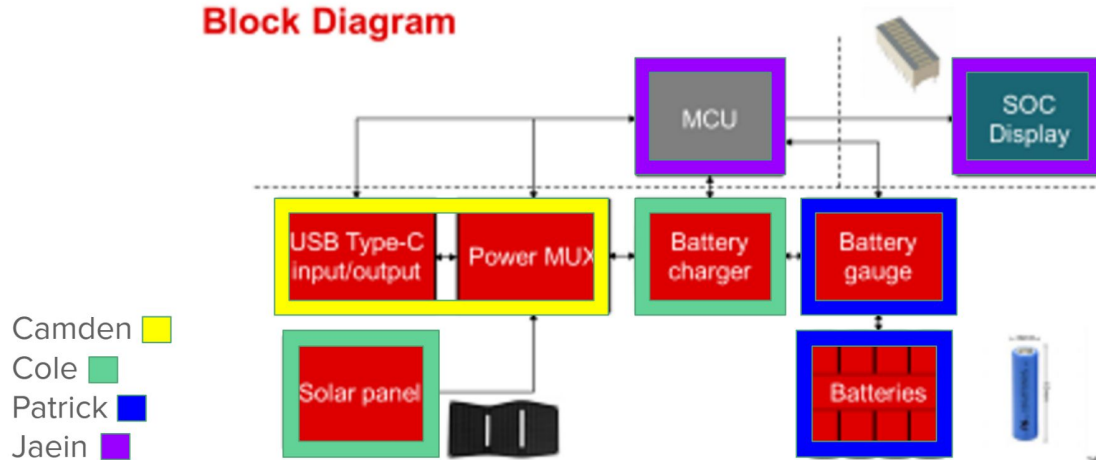
Project Summary

- Natural disasters can cause unexpected power outages resulting in the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.



Project/Subsystem Overview

Block Diagram





Project Timeline



January

Individual Subsystem Finalization

Wrap up subsystem design and
validation testing



February

Integration Design

Work together to integrate the
designs of each subsystem



March

Assembly/Validation

Assemble the final pcb and 3D
printed enclosure. Perform
validation testing and identify
problems that occur



April

Validation/Debugging

Engineer a solution to problems
that arise as a result of
integration

Patrick Wang (Battery Gauge)

Accomplishments since last update 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Ran tests with battery gauge EVM and tested CUV and COV faults • Able to open CHG and DSG FETs to enable power path and get current and voltage to flow through on my board • Integrated with the power supply to charge battery pack,	• Currently board has an undervoltage fault, preventing further integration with the charger • Clear fault by this next meeting, plug into charger



Patrick Wang

Dashboard

Auto Refresh is OFF - Click to Turn On

bqStudio Version: 1.3.124

EV2400
Version: 0.32

I2C

bq28z620R1
2620_1_00
Addr: 0xAA
-273.2 degC

3991 mV
100%

500
1000
2000
-61

Registers

Registers

Name	Value	Units
Manufacturer Access	0x0076	hex
At Rate	0	mA
At Rate Time To Empty	65535	min
Temperature	-273.2	degC
Voltage	3991	mV
Current	-62	mA
Max Error	100	%
Average Current	-53	mA
Average Power	-21	cW
BTP Dtg Sst	-1	mAh
BTP Chg Sst	-1	mAh
Relative State of Charge	100	%
Remaining Capacity	0	mAh
Full charge Capacity	0	mAh
Average Time to Empty	0	min

Name	Value	Units
Max Load Time to Empty	0	min
Charging Current	0	mA
Charging Voltage	0	mV
Cycle Count	0	-
State of Health	0	%
Cell 1 Voltage	3991	mV
Cell 2 Voltage	0	mV
BAT pin voltage	3993	mV
PACOK pin voltage	4063	mV
Cell 1 Current	-58	mA
Cell 2 Current	0	mA
Cell 1 Power	-23	cW
Cell 2 Power	0	cW
Power	-24	cW
Int Temperature	21.0	degC

Name	Value	Units
Fit Full Chg Q	0	mAh
Fit Full Chg E	0	cWh
True Rem Q	-3	mAh
True Rem E	-1	cWh
Initial Q	0	mAh
Initial E	0	cWh
True Full Chg Q	0	mAh
True Full Chg E	0	cWh
T_amb	-273.2	degC
T_ambient	-273.2	degC
Cell 1 RaScale	1000	-
Cell 2 RaScale	0	-
Cell 1 CompRes	0	m...
Cell 2 CompRes	0	m...
PackGrid	0	-

Name	Value	Units
Cell 2 DOD0	0	-
DOD0 Passed Q	3	mAh
DOD0 Passed E	1	cWh
DOD0 Time	0	h/16
Cell 1 DODEOC	5712	-
Cell 2 DODEOC	0	-
Cell 1 QMax	4400	mAh
Cell 2 QMax	4400	mAh
Cell 1 QMax DOD0	0	-
Cell 2 QMax DOD0	0	-
QMax Passed Q	3	mAh
QMax Time	0	h/16
Temp k	1.0	-
Temp a	1000	-
Cell 1 Raw DOD	0	-

Start Log Scan Refresh

Commands

HW_VERSION
FW_VERSION
FW_BUILD
CHEM_ID
SHUTDOWN
CC_OFFSET
CHG_FET_TOGGLE
DSG_FET_TOGGLE
GAUGE_EN
FET_EN
LIFETIME_EN
LT_RESET
LT_FLUSH
PF_EN
PF_CLEAR
CAL_TOGGLE
RESET
SHOOT_CALIB

Log Panel Clear Log

Transaction Log

Name	Cmd	Result	Read ...
FET_EN	0x22	N/A	N/A
CHG...	0x1F	N/A	N/A
DSG...	0x20	N/A	N/A

TEXAS INSTRUMENTS

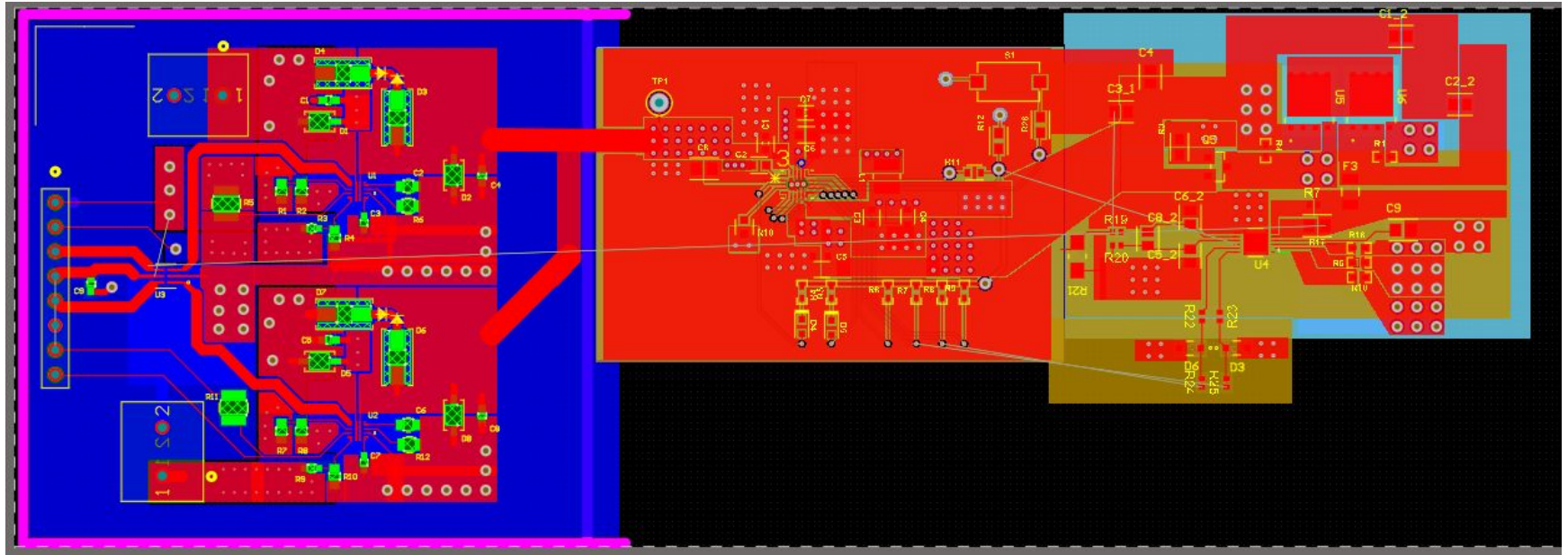


Cole Miller (Battery Charger)

Accomplishments since last update 35 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Integration Testing with Power MUX • Successfully Integrate with EV2400 • Created final integrated Board • Helped troubleshoot battery Gauge	• Order Final integrated PCB • Battery Gauge Integration • Quiescent current calculations • Power Calculations

Cole Miller

- Board is currently 6.325in by 2.2in. The FSR mandated the size of the board to be below 8.8in by 5.85in. Leaving space for the walls of the battery pack and fasteners.

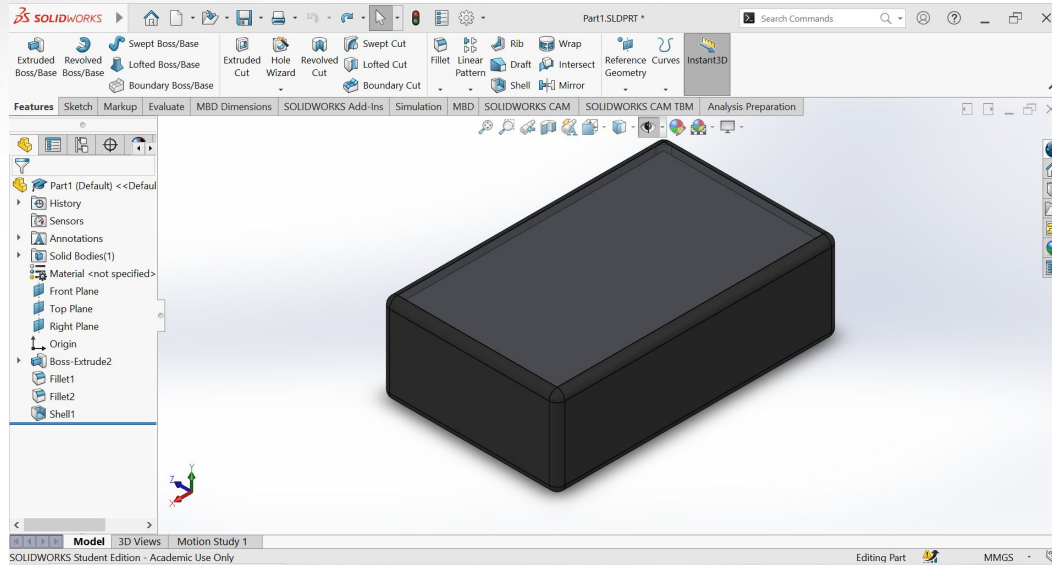




Camden Pomeroy (Power MUX)

Accomplishments since last update 20 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Integration testing with Battery Charger • Created Preliminary battery pack design using Solidworks software • Design Review Integrated Schematic and PCB layout design	• Finalize Solidworks Design • Submit Prototyping Request through FEDC • Full Integration testing with Patrick's (Battery Gauge) and Jaein's (MCU) boards

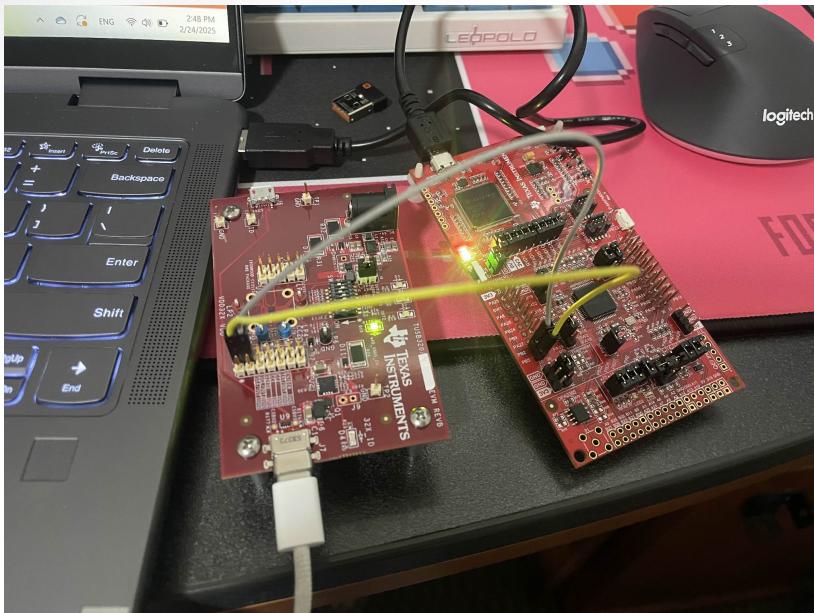
Camden Pomeroy



Jaein Cha (MCU)

Accomplishments since last update 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Software integration test with TUSB320 EVM for I2C read/write • Updated MCU & TUSB320 schematics for full integration with other subsystems (including package replacement) • Resolved the debugger-on board MCU connection issue	• I2C debugging (problem during data transfer) • Test I2C read/write with other subsystem EVMs and prototype boards • Integrate updated MCU PCB layout to the integrated board layout (currently finalizing updated PCB layout) • Implement MPPT for efficient power transfer

Jaein Cha



Execution & Validation Plan

[illegible]



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Questions?